
MACQUARIE
University
DATA
VISUALISATION

MACQUARIE University

Reidar Lystad



Bianca Casimiro



Thomas Fung




AI-generated image

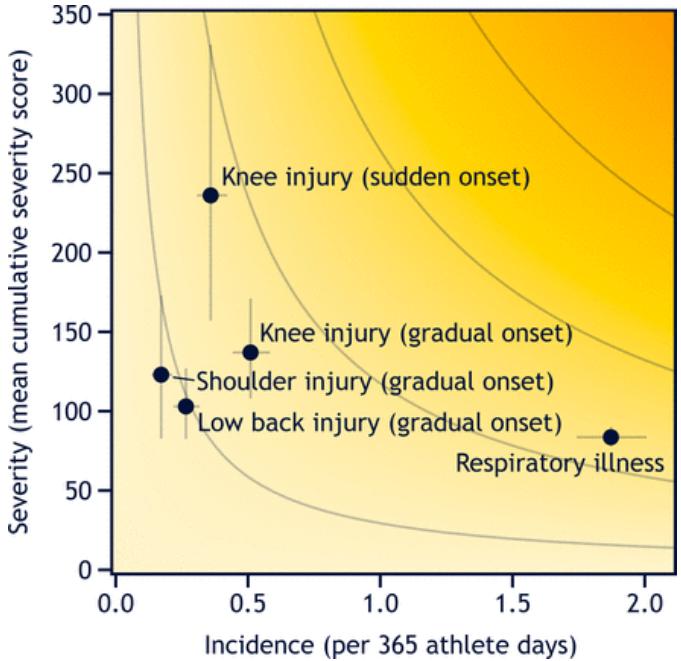
British Journal of
Sports Medicine

International Olympic Committee consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including STROBE Extension for Sport Injury and Illness Surveillance (STROBE-SIIS))

Roald Bahr ,[1,2] Ben Clarsen ,[1,3] Wayne Derman,[4] Jiri Dvorak,[5] Carolyn A Emery ,[6,7] Caroline F Finch ,[8] Martin Hägglund ,[9] Astrid Junge,[10,11] Simon Kemp,[12,13] Karim M Khan ,[14,15] Stephen W Marshall,[16] Willem Meeuwisse,[17,18] Margo Mountjoy ,[19,20] John W Orchard ,[21] Babette Pluim,[22,23,24] Kenneth L Quarrie ,[25,26] Bruce Reider,[27] Martin Schwellnus,[28] Torbjørn Soligard ,[29,30] Keith A Stokes ,[31,32] Toomas Timpka ,[33,34] Evert Verhagen ,[35] Abhinav Bindra,[36] Richard Budgett,[29] Lars Engebretsen,[1,29] Uğur Erdener,[29] Karim Chamari[37]

EXAMPLE

Burden by body area (and mode of onset)



Bahr et al (2020)

EXAMPLE

Injuries by body regions/areas and type of tissue/pathology

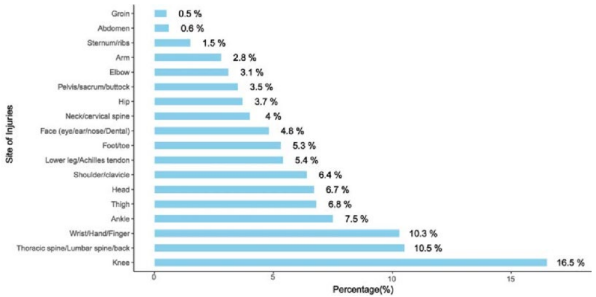
Body region	Body area	N (%)
Head and neck	Head	
	Neck	
Upper limb	Shoulder	
	Upper arm	
	Elbow	
	Forearm	
	Wrist	
	Hand	
Trunk	Chest	
	Thoracic spine	
	Lumbosacral	
	Abdomen	
Lower limb	Hip/groin	
	Thigh	
	Knee	
	Lower leg	
	Ankle	
	Foot	
Unspecified	Region unspecified	

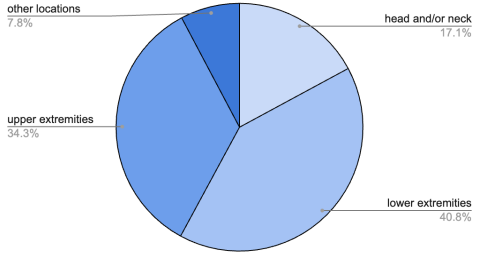
Tissue	Pathology type	N (%)
Muscle/Tendon	Muscle injury	
	Muscle contusion	
	Muscle compartment syndrome	
	Tendinopathy	
	Tendon rupture	
Nervous	Brain/Spinal cord injury	
	Peripheral nerve Injury	
Bone	Fracture	
	Bone stress injury	
	Bone contusion	
	Avascular necrosis	
	Physis injury	
Cartilage/Synovium/Bursa	Cartilage injury	
	Arthritis	
	Synovitis/Capsulitis	
	Bursitis	
Ligament/Joint capsule	Joint sprain	
	Chronic instability	
Superficial tissues/skin	Contusion (superficial)	
	Laceration	
	Abrasion	
Vessels	Vascular trauma	
Stump	Stump injury	
Internal organs	Organ trauma	
Non-specific	Injury without tissue type specified	

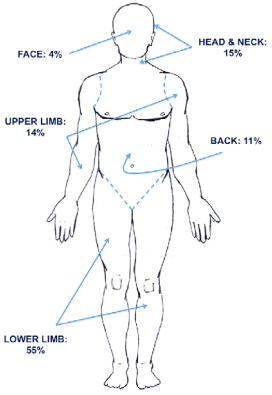
Bahr et al (2020)

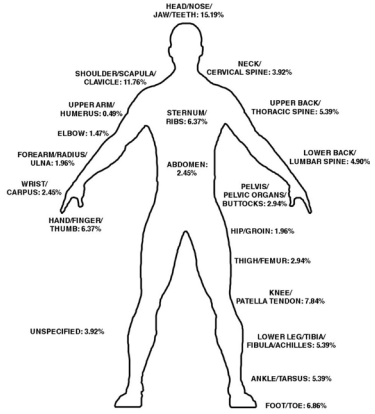
EXAMPLE

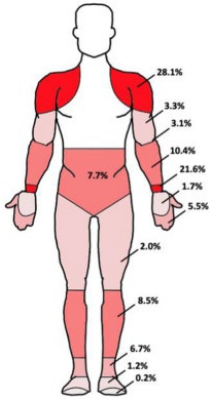
Injuries by body regions/areas

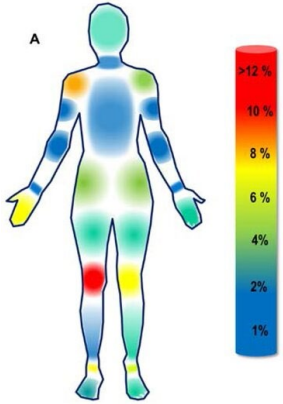








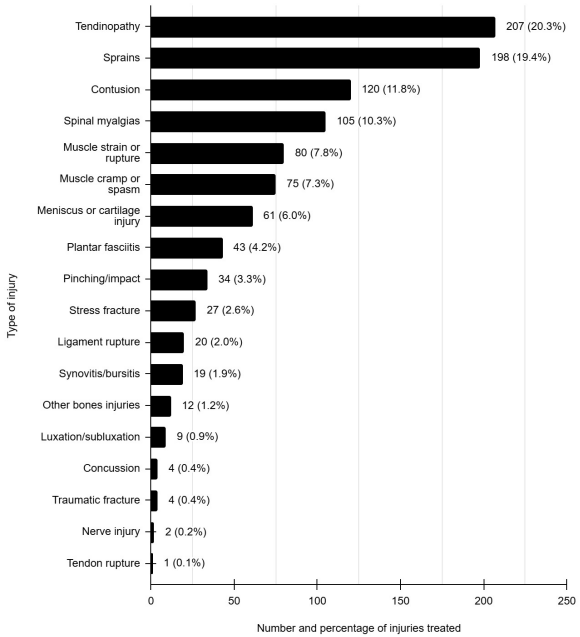


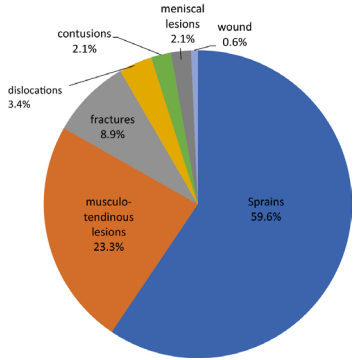


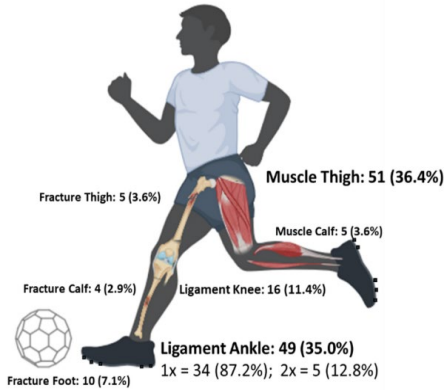
Zhang et al (2023); Jandhyala et al (2024); Jordan et al (2019); Ryan et al (2020); Hurt et al (2023); Sole et al (2017)

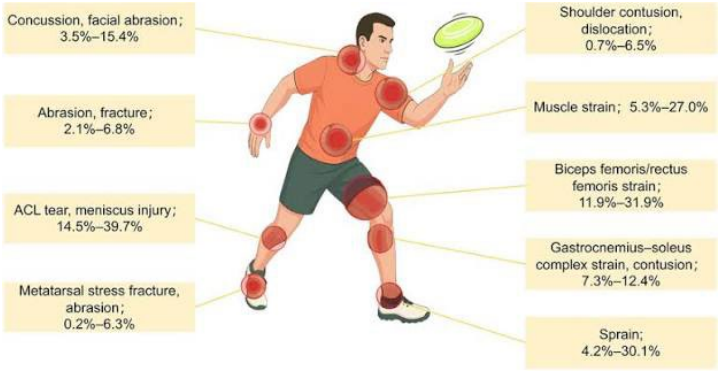
EXAMPLE

Injuries by type of tissue/pathology









Rocco Siqueira et al (2025); Saragaglia et al (2023); Schultz et al (2025); Chen et al (2026)

JSAMS
Journal of Science and Medicine in Sport

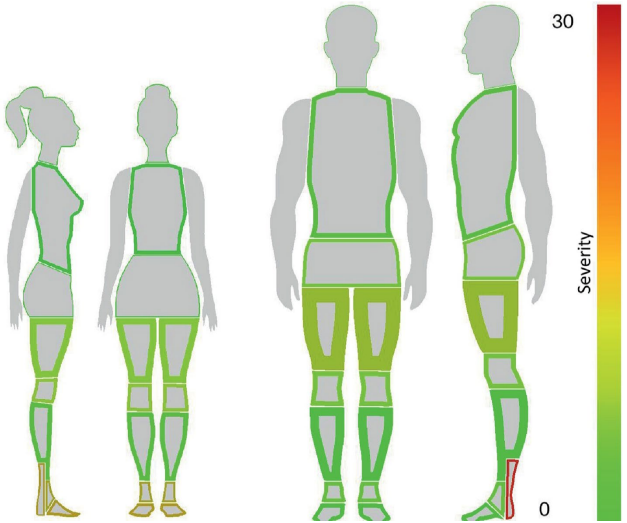
Data visualisation for reporting epidemiological data on injury and other health conditions in sport

Pascal Edouard [a b c], Spyridon Iatropoulos [a f], Pierre-Eddy Dandrieux [a b f], Karsten Hollander [d], Astrid Junge [d], Laurent Navarro [e]



EXAMPLE

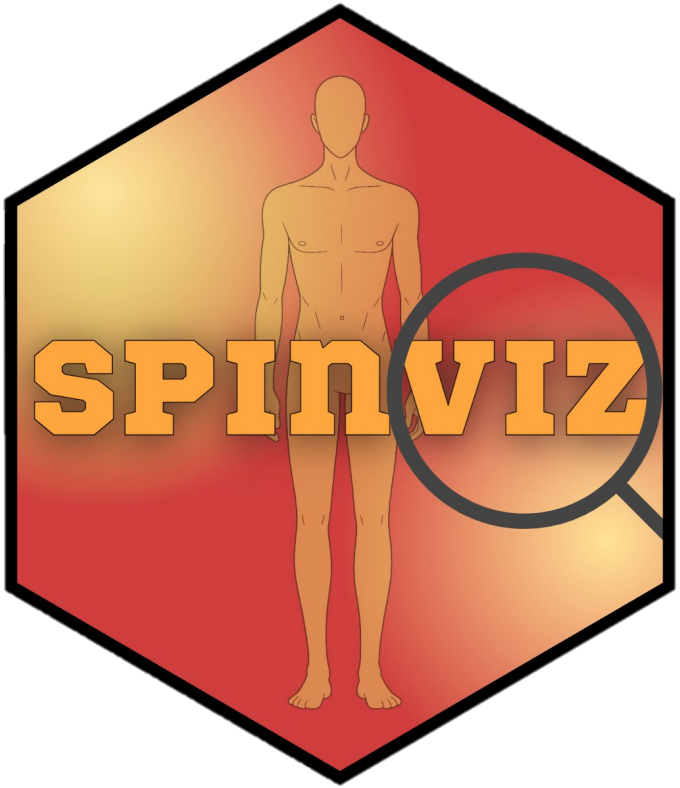
Proposed data visualisation for reporting of epidemiological data on injuries in sport

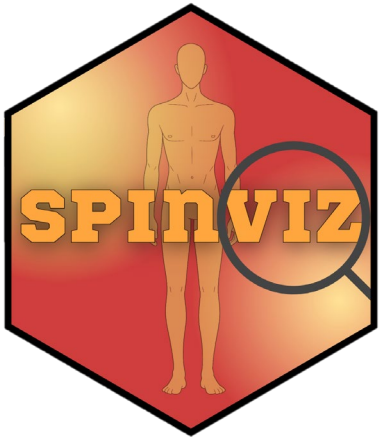


Edouard et al (2026)

“There is no such thing as information overload. There is only bad design.

~ Edward Tufte


SPINVIZ

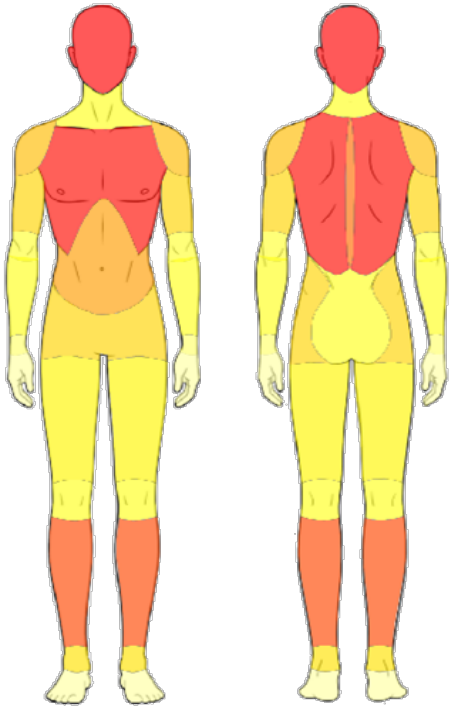


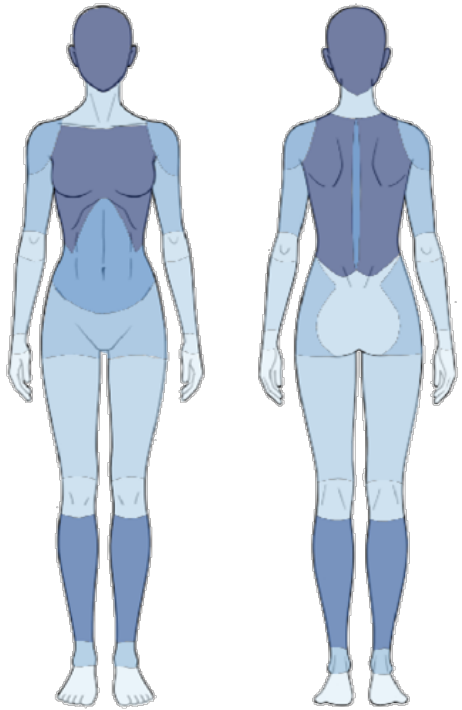
1 Heatmap plot
- Silhouette gender
- Anatomical view
- Anatomical granularity
- Anatomical labelling
- Colour palette

2 Sunburst plot
- Granularity
- Labelling
- Colour palette

EXAMPLE

Heatmap plot



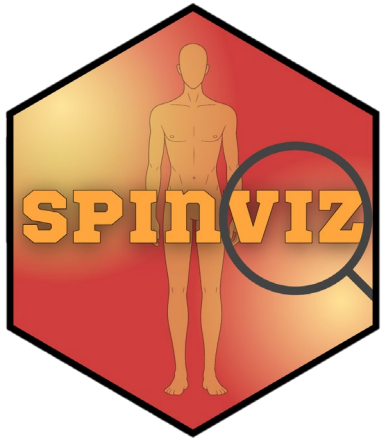


EXAMPLE

Sunburst plot



LIVE DEMO



1

Posit Cloud

Sign up for a free Posit Cloud account
or log in using your existing Google
or GitHub account credentials.

2





Retrieve copy of the workshop R script:

```
download.file(
        url = "https://tinyurl.com/spinviz",
        destfile = "spinviz_workshop.R")
```

Thank you!

